Final Project

Inferential Statistics 2025/2026

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1 Abstract

This report analyzes the determinants of academic performance using the UCI "Student Performance Data Set," derived from Mathematics classes in two Portuguese schools (Gabriel Pereira and Mousinho da Silveira) and collected through school reports and questionnaires. The study aims to identify the demographic, social, and behavioral factors driving final grades (G3), using empirical evidence to validate or refuse common assumptions about student performance. Employing Chi-Squared tests, along with parametric tests and regression analysis, we investigate variables ranging from family background to social habits.

2 Introduction

3 Data description and exploration

4 Statistical Analysis

4.1 T-Test (Group Comparison)

In this section, we compare the mean final grade (G3) between two different categories.

4.1.1 Hypothesis A (Technological)

Is there a significant difference in grades between those who have Internet at home and those who do not?

```r
t.test(G3 ~ internet, data = data)
```

```
##
##  Welch Two Sample t-test
##
## data:  G3 by internet
## t = -1.9894, df = 94.219, p-value = 0.04955
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
##  -2.413445670 -0.002415065
## sample estimates:
##  mean in group no mean in group yes
##          9.409091         10.617021
```

The results indicate that students with Internet access at home have a higher mean grade compared to those without. The T-test yields a p-value slightly above the 0.05 threshold or right on the margin depending on the variance assumption, but generally suggests a positive tendency. Specifically, the mean grade for students with Internet is approximately **10.6**, while for those without, it is **9.4**.

This difference might stem from easier access to study materials, online exercises, and educational platforms. However, since the access to the internet nowadays is ubiquitous, the "no" group might represent a specific socio-economic demographic that faces other challenges affecting their performance.

4.1.2 Hypothesis B (Sentimental)

Do students in a romantic relationship have lower grades compared to single students?

```
t.test(G3 ~ romantic, data = data)
```

```
##
##  Welch Two Sample t-test
##
## data:  G3 by romantic
## t = 2.5122, df = 240.07, p-value = 0.01266
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
##  0.2721553 2.2493333
## sample estimates:
##  mean in group no mean in group yes
##         10.836502          9.575758
```

Here we observe a statistically significant difference. The mean grade for single students is approximately **10.8**, whereas students in a romantic relationship have a mean grade of **9.6**. The p-value confirms that this difference is unlikely to be due to chance.

This finding aligns with the "time allocation" theory: maintaining a romantic relationship requires time and emotional energy that might otherwise be dedicated to studying. The observed drop of more than 1 grade point is a substantial impact on the final performance.